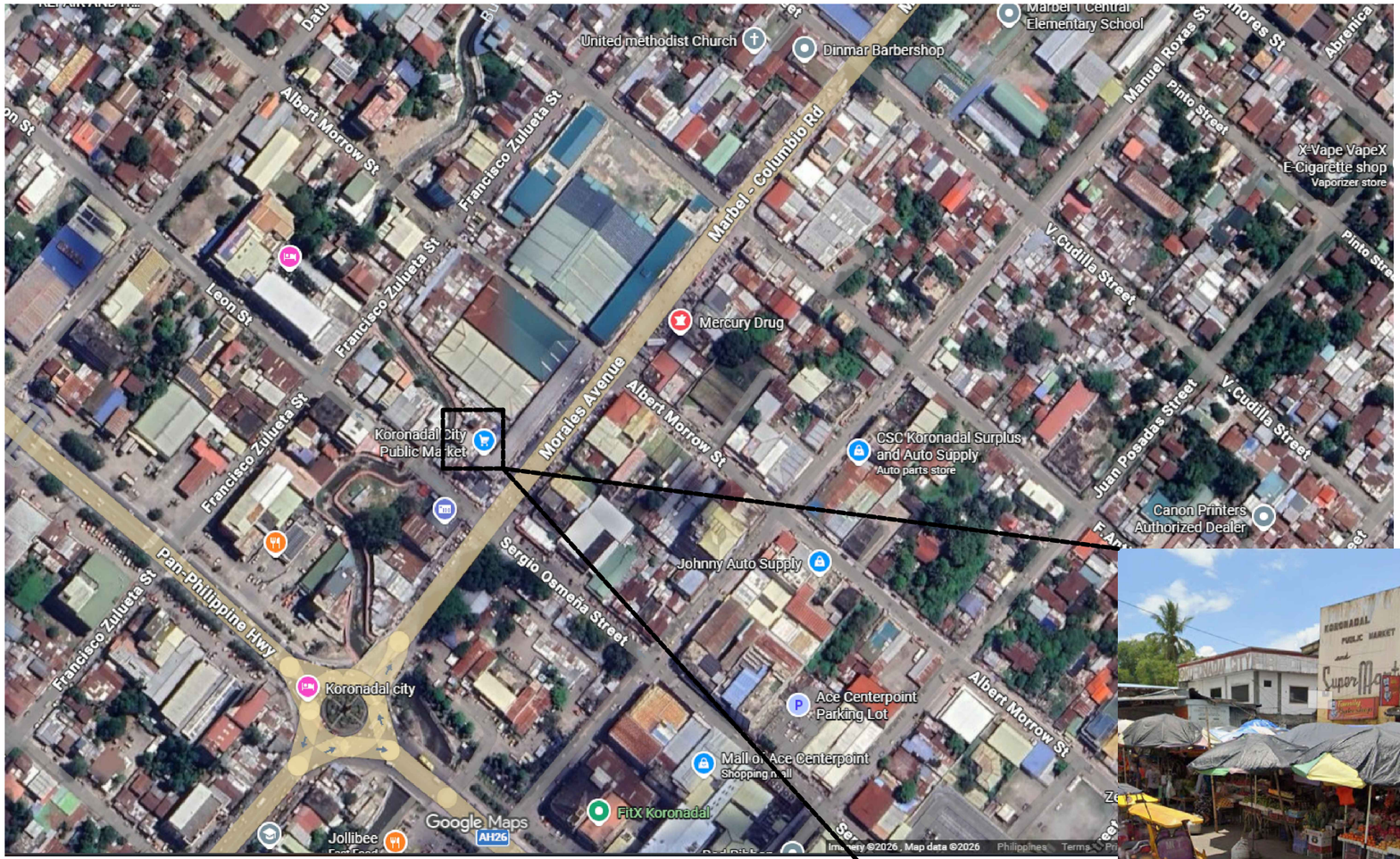
REPUBLIC OF THE PHILIPPINES
PROVINCE OF SOUTH COTABATO
CITY OF KORONADAL

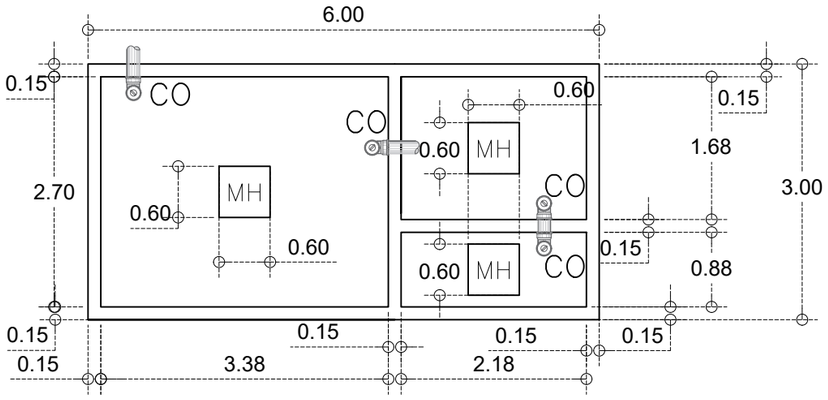
CONSTRUCTION OF SEPTIC TANK

KORONADAL PUBLIC MARKET
BRGY.ZONE I, CITY OF KORONADAL



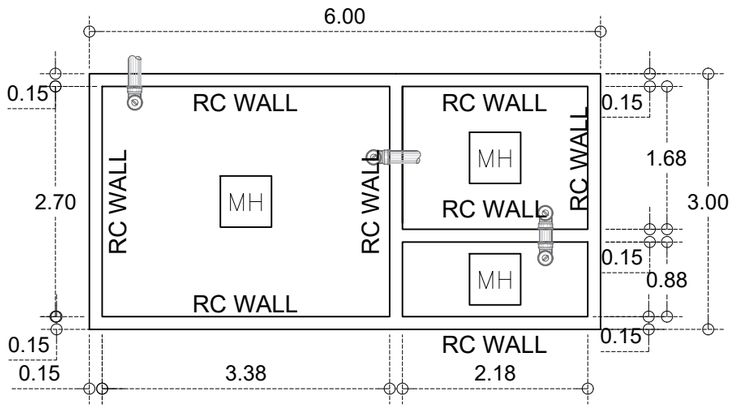
VICINITY MAP
NTS

	PREPARED BY:	CHECKED BY:	REVIEWED BY:	PROJECT TITLE:	RECOMMENDING APPROVAL:	APPROVED BY:	SHEET NO:
REPUBLIC OF THE PHILIPPINES PROVINCE OF SOUTH COTABATO CITY OF KORONADAL BDP	ENGR. SEVERINO JOHNROME E. USERO	ENGR. JAYSON CANTOR	ENGR. JOHN JOVI Z. ANGOR	CONSTRUCTION SEPTIC TANK	ENGR. RUEL C. TIANCHON, MPA, REA, REB	ATTY. ELIORDO U. OGENA	P-01
	ENGINEER-1	ENGINEER-1	ENGINEER- IV/BDP COORDINATOR	LOCATION: CITY OF KORONADAL	CITY ENGINEER	CITY MAYOR	03



SEPTIC VAULT

TOP VIEW
SCALE 1: 50 MTS.



SEPTIC VAULT

TOP VIEW
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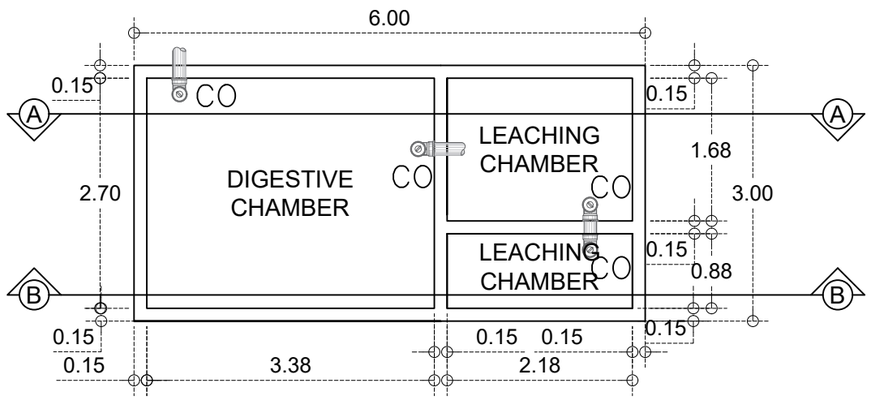
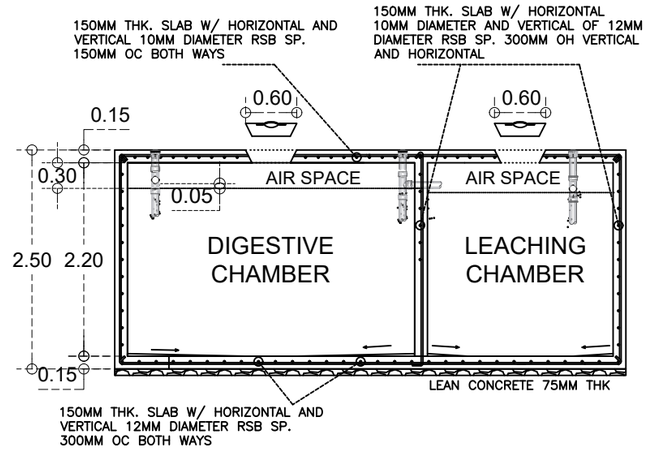
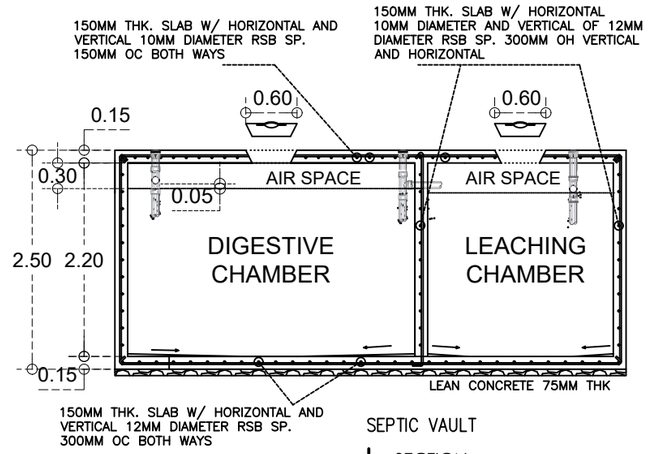


SEPTIC VAULT

DIAGRAM
SCALE 1: 50 MTS.





NOTES:
1. ADD EXTRA BARS AROUND MANHOLE

SEPTIC VAULT

SECTION
SCALE 1: 50 MTS

REPUBLIC OF THE PHILIPPINES PROVINCE OF SOUTH COTABATO CITY OF KORONADAL BDP	PREPARED BY:	CHECKED BY:	REVIEWED BY:	PROJECT TITLE:	RECOMMENDING APPROVAL:	APPROVED BY:	SHEET NO:
				CONSTRUCTION SEPTIC TANK			P-02
	ENGR. SEVERINO JOHNROME E. USERO	ENGR. JAYSON CANTOR	ENGR. JOHN JOVI Z. ANGOR		ENGR. RUEL C. TIANCHON, MPA, REA, REB	ATTY. ELIORDO U. OGENA	03
	ENGINEER- I	ENGINEER-I	ENGINEER- IV/DDP COORDINATOR	LOCATION: CITY OF KORONADAL	CITY ENGINEER	CITY MAYOR	

PLUMBING NOTES:

1. GRADES OF HORIZONTAL PIPINGS
   RUN ALL HORIZONTAL PIPINGS IN PERFECT ALIGNMENT AND AT A FORM GRADE NOT LESS THAN TWO PERCENT (2%)
2. CHANGE IN DIRECTION
   ALL CHANGE IN DIRECTION SHALL BE MADE BY APPROPRIATE USE OF FORTY–FIVE DEGREES (45˚) WYES, LONG SWEEP QUARTER BEND, SIXTH–EIIGHT OR SIXTEENTH BEND. WHEN THE CHANGE OF FLOW IS FROM HORIZONTAL TO VERTICAL, A SINGLE 1/8 BEND COMBINATION MAY BE USED ON VERTICAL STACKS, AND SHORT QUARTER BENDS MAYBE USED ON WASTE LINE, TEE AND CROSSES MAYBE USED IN BENT PIPES.
3. PROHIBITED FITTINGS
   NO DOUBLE HUB OR TEE BRANCH SHALL BE USED ON HORIZONTAL SOIL AND WASTE LINES, THE DRILLINGS AND TAPPING OF HOUSE DRAIN, WASTE OR VENT PIPES AND USE OF SADDLE HUB AND BEND ARE PROHIBITED.
4. PIPE CLEAN–OUTS
   CLEAN–OUTS ARE REQUIRED UNDER THE FOLLOWING CONDITIONS:
   a) EVERY CHANGE OF HORIZONTAL DIRECTION EXCEEDING TWENTY TWO AND ONE–HALF DEGREES (22 1/2˚).
   b) ONE AND ONE–HALF METERS (1.50m.) INSIDE THE PROPERTY LINES BEFORE THE HOUSE DRAINAGE CONNECTION.
   c) EVERY FIFTEEN METERS (15.00m) IN HORIZONTAL RUN OF PIPES.
   d) AT THE END OF ANY HORIZONTAL PIPE LINES.
5. THE DIGESTION CHAMBER OF SEPTIC VAULT MUST BE WATERPROOFED.
6. NOT LESS THAN 0.30 METER OF AIR SPACE MUST BE LEFT BETWEEN THE TOP OF THE SEWAGE AND THE UNDER PART OF VAULT ROOF SLAB.
7. NO SEPTIC VAULT SHALL BE CONSTRUCTED UNDER THE BUILDING.
8. ALL PLUMBING WORKS SHALL BE UNDER THE SUPERVISION OF A LICENSED MASTER PLUMBER AND A LICENSED PLUMBING CONTRACTOR.

DESIGN CRITERIA

DESIGN CRITERIA

(Septic Tank – Public Market Comfort Room)

i. DESIGN USE
Public Market Comfort Room (Male & Female Toilets and Floor Drains Only)

ii. DESIGN LIFE
Minimum of 25 years

iii. LIVE LOAD (Cover Slab – Foot Traffic Only)
Live Load = 3.0 kPa (300 kg/m²)

iv. ALLOWABLE STRESSES

Concrete
a. For bottom slab, walls, and top slab
$f'c$ = 20 MPa

Concrete Masonry Units (if applicable)
$f'm$ = 6.90 MPa
τm = 2.41 MPa

Reinforcing Steel Bars
f_y = 275 MPa (Grade 40 deformed bars)
Minimum bar diameter = 10 mm

v. ASSUMED SOIL BEARING CAPACITY
q_a = 100 kPa

vi. GROUND WATER CONDITION
Groundwater table assumed 1500 BELOW GROUND LEVEL

vii. DESIGN FLOW CONSIDERATION
- Wastewater from toilets and floor drains only
- No rainwater or storm drainage connected
- Peak flow considered for public use

viii. PIPE DESIGN CRITERIA
- Toilet and floor drain lines: 100 mm Ø
- Main vertical stack to septic tank: 150 mm Ø
- Minimum horizontal slope: 2% (1:50)
- Pipes designed for gravity flow

ix. SEPTIC TANK CONFIGURATION
- Multi–chamber septic tank system
  - Digestive Chamber
  - Leaching Chamber
- Freeboard provided: 300 mm
- Liquid depth: 2.0 m

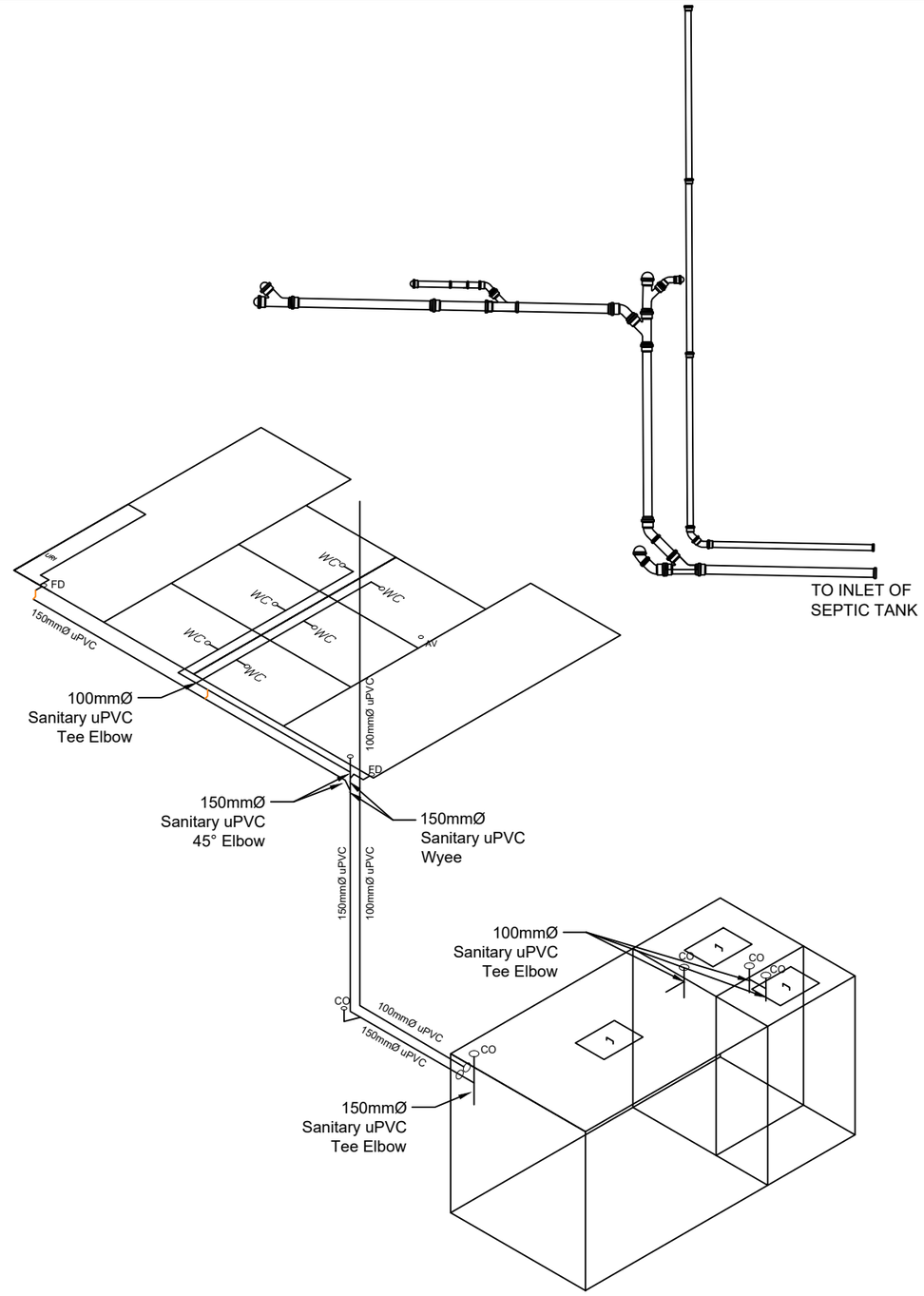
x. STRUCTURAL ASSUMPTIONS
- Septic tank buried below finished ground level
- No vehicular load above septic tank
- Adequate ventilation provided via vent pipes

LEGEND:

AC	AIR CHAMBER
PT	BLADDER/PNEUMATIC TANK
E	ELECTRICMOTOR
P	PUMP
U	UNION PATENTE
FTV	FOOT VALVE
FV	FLOAT VALVE
PP	PRIMING PORT
CB	CATCH BASIN
CDP	CONCRETE DRAIN PIPE
CV	CHECK VALVE
HB	HOSE BIBB
FD	FLOOR DRAIN
PPR	POLYPROPYLENE COLD WATER LINE
PPR	POLYPROPYLENE COLD WATER RISER
GV	GATE VALVE
MH	MANHOLE
PPR CWL	POLYPYRENE COLD WATER LINE
PVCDS	POLYVINYL CHLORIDE DOWNSPOUT
PVCDP	POLYVINYL CHLORIDE DRAIN PIPE
PVCDS	POLYVINYL CHLORIDE DOWNSPOUT
CO	CLEAN OUT
PVCSP	POLYVINYL CHLORIDE SOIL PIPE
PVCVAC	POLYVINYL CHLORIDE VENT. ACROSS CEILING
PVCVP	POLYVINYL CHLORIDE VENT. PIPE
PVCVS	POLYVINYL CHLORIDE VENT STACK
PVCVTR/W	POLYVINYL CHLORIDE VENT TROUGH ROOF/WA
PVCWP	POLYVINYL CHLORIDE WASTE PIPE
PVCWS	POLYVINYL CHLORIDE WASTE STACK
RD	ROOF DRAIN
UD	URINAL DRAIN
WC	WATER CLOSET
LAV	LAVATORY
KS	KITCHEN SINK

NOTE:

- Septic tank shall comply with the National Plumbing Code of the Philippines (NPCP) and DPWH standards.
- Proper maintenance access (manholes and clean–outs) shall be provided.
- All materials shall be new and of approved quality.



SEPTIC VAULT
ISOMETRIC DIAGRAM
SCALE 1: 50 MTS

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